Using a novel light field measurement system to simultaneously measure 3D-3C velocity fields and 3D surface profiles

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Using a novel light field measurement system to simultaneously measure 3D-3C velocity fields and 3D surface profiles

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Introduction

Many different systems are known that are able to volumetrically measure all three components of a velocity field based on classical particle image velocimetry (PIV). Generally, these systems consist of several detectors, i.e. cameras that look onto the volume from different directions. A defocused image of a seeding particle is thereby captured by several detectors from which the position of the seeding particle can thus be determined. If one also needs to measure the surface structure of solid bodies, an additional measurement system is required.
Within the context of a research project that aims to analyze the burial of objects at the bottom of the ocean (Menzel, 2013), the flow around a bedded cylinder on a sand bed at the bottom of a measurement channel is to be analyzed and the subsequent burial of the cylinder described. By doing so, it is possible to analyze the change of the flow structure due to the surface profile of the sand bed, which in turn is also dependent on the flow. For this, a novel light field measurement system was used, which is able to measure all three components of the three dimensional velocity field, as well as a three dimensional surface profile, with just one sensor.

Measurement principle

The basic principles of so-called light field cameras were already described by Ives (1903) and Lippman (1903). They placed several small lenses in front of a detector, thus creating a multiple image (light field) which enables a view of the scene from different angles. This results in a high depth of focus, as well as the capability to retroactively change the focus of the image. Furthermore, analogous to a multi-stereo system, it is also possible to extract information regarding the 3D structure of the photographed surface.
During the following centuries, the technology was further investigated. Ives (1928) added a large principal lens to the setup, which produces a reduced image of the area. This image is then projected onto a detector through a field of pinhole cameras, thus allowing for an exposure of scenes which are considerably larger than the camera field itself. At the beginning of the 21st century, customized models, which were then called plenoptic cameras, were created for various applications. Levoy et al. (2006) created a plenoptic microscope which, instead of using the principal lens to produce a reduced image, used microscope optics to enlarge the scene.
The systems now usually consist of a conventional digital camera, which is enhanced by a microlens array (MLA). As shown in Fig. 1, the principal lens creates virtual images of the objects. Each microlens then creates an image of the scene, which is projected onto the

camera sensor. The resulting light field image is subsequently used to create either 2D images with a desired focal plane or 3D reconstructions. Further information on the temporal development can be found in Roberts (2003).

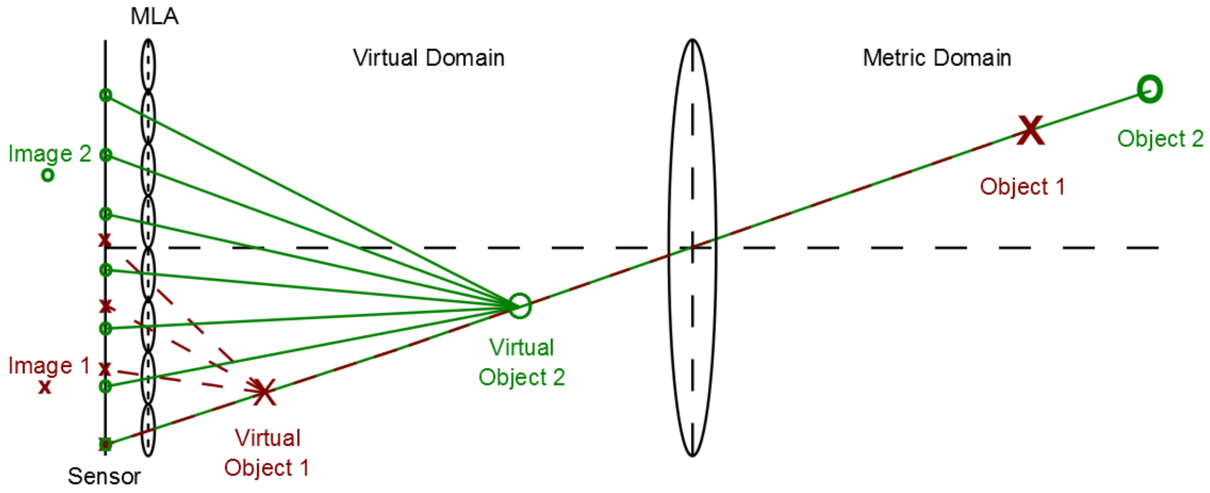


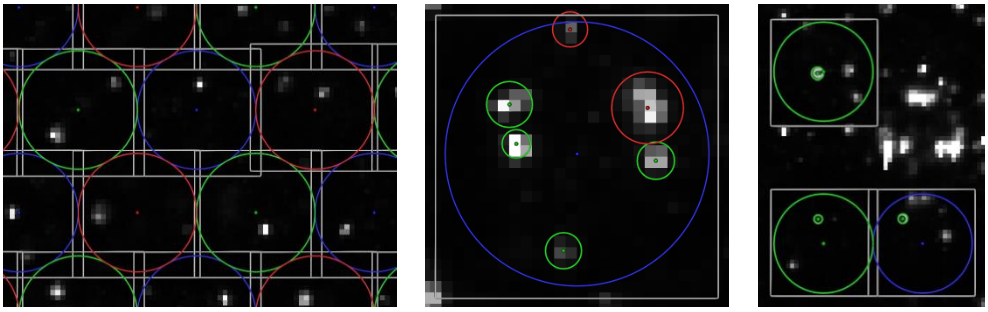
Fig. 1: Schematic setup of a plenoptic camera.

The camera which is used for the underlying work was introduced by Perwass & Wietzke (2012). It has an enhanced depth resolution due to the use of microlenses with varying focal lengths. Thus, one can cover a wider depth range without having to reduce the resolution.

Particle tracking with plenoptic cameras

The presented method of flow estimation is based on the visual tracking of particles which are subjected to the flow. Thanks to the plenoptic camera that was used, it is not only possible to measure the movement parallel to the image sensor, but also the movement towards and away from the camera. From this, the full 3D trajectories of all detected particles can be estimated. As can be seen in Fig. 2, one can also obtain the positions of those particles that would not be detectable with a stereo camera system due to overlapping: if two particles overlap in one microlens image, then they will be separated and therefore be detectable in other microlens images.
In order to resolve high velocities, the camera can also be used in a double image mode. In that case, two images are captured within a very short time interval. The resulting trajectories are then based on only two points respectively.
The algorithm for the flow estimation is made up of several stages. To begin with, the image sequence is scanned for particles in the microlens images, which are then classified as either invalid or valid. Fig. 2b shows the valid particles marked by a green circle and the invalid particles marked by a red circle. For the next step, the displacement of the particles between the respective double images is computed. The spatial relations can be obtained via the detected particles in the various microlens images. Fig. 2a, for example, shows the lenses of varying focal lengths in different colors respectively. The 3D positions can then be triangulated.
This ultimately results in many individual 3D positions with associated position changes, which correspond to the local velocities. In the post processing step, the vector field is interpolated onto a uniform grid, thereby simplifying the following processing operations. If the flow field is assumed to be steady state, then the grid can additionally be averaged over the entire sequence, which further enhances the accuracy.
This method of using a plenoptic camera for particle tracking offers several advantages over other methods. Compared to multi camera systems, no time-consuming external calibration (between the cameras) is needed. Thanks to the direct computation of the local velocities

when using the plenoptic method, the properties of the grid for the statistical average can be optimally chosen during post processing.



a b c

Fig. 2: Particle tracking with a plenoptic camera.

A further significant improvement compared to other optical methods is the fact that the data from several microlenses are used for one single particle. Up to 12 or more lenses are used for the estimate of a single 3D position. Fig. 2c, for example, shows the assignment of particles from three different microlenses to a single 3D particle. In contrast, stereo or tri-stereo camera systems can only use two or three images respectively. Due to this high redundancy, it is possible to obtain a valid temporal tracking and a robust 3D estimate even with a very high particle density.

Experimental setup

In order to demonstrate the capabilities of the presented measurement system, an existing experimental setup at the Department of Fluid Mechanics was used, that enables the analysis of the burial of objects at the bottom of the ocean. A cylinder with a diameter of D = 58 mm and a length of L = 190 mm was placed on a sand bed with grain diameters of 0.1 mm < d_K < 0.3 mm and water was streamed perpendicular to the cylinder axis. Fig. 3 shows the measurement channel with its relevant dimensions, as well as the experimental setup. The flow structures and the surface profile of the resulting sand bed were previously published in Menzel et al. (2013) and Menzel & Leder (2013).
The following experiments were conducted around the cylinder after it had reached a silted state. Therefore, scour structures are clearly detectable. The surface profile was measured with the light filed camera by taking individual photographs. For this, coaxial lighting conditions were realized with the help of several spotlights, so as to create a homogeneous illumination of the photographed surface without any shadows. As a comparison, the surface profile was also measured with a laser distance sensor. Thereafter, the channel was carefully filled with water. The averaged flow structure was then measured with the use of a 120 mJ Nd:YAG double pulse laser. Polyamide particles with a diameter of roughly d_p = 54 µm were used as seeding particles and a particle density of about 10% of a correspondent classic PIV measurement was choosen. A camera lens with a focal length of 100 mm was provided by the measurement system, which, together with the measuring distance due to the experimental setup, allowed for a measuring volume of about 100 x 100 x 60 mm³, though larger volumes could also have been obtained with the given illumination. A frame rate of 4 Hz was obtained with the 29 megapixel camera, with the images being captured within real time. The 3D surface profiles could also be visualized within real time.

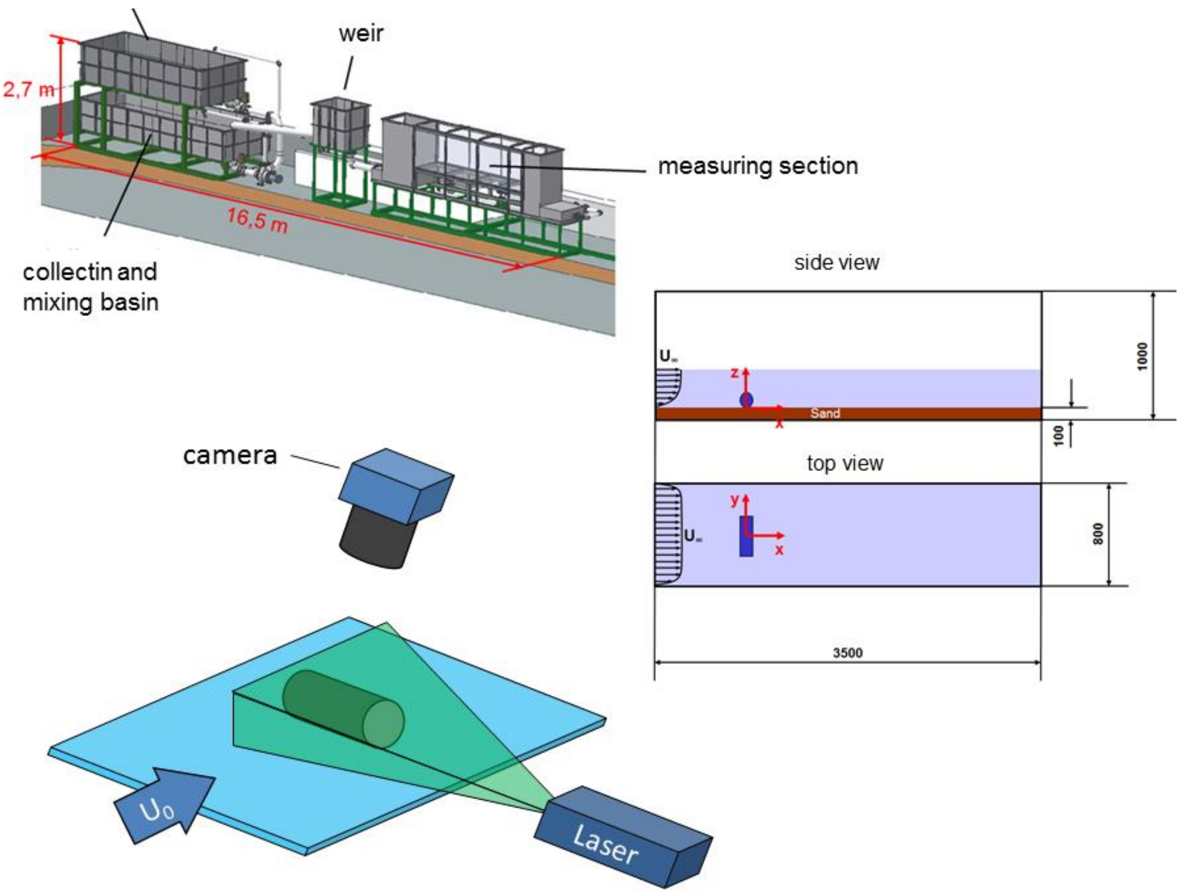


Fig. 3: Measurement channel, setup and coordinate system.

As previously mentioned, a laser distance sensor with automatic traversing was used in order to validate the measured surface profiles. The sensor uses triangulation to obtain an absolute accuracy for the measured distance of 0.3 mm, though the measurement point itself has a diameter of 2.3 mm due to the expansion of the laser beam. The resulting spatial resolution is thus roughly 2 mm in the plane perpendicular to the laser distance sensor and about 1 mm in the depth, though areas of steep elevation changes are subjected to higher inaccuracies.

Measurement of the surface profile and validation

The surface profile of almost the entire channel was measured with the laser distance sensor. The area around the bedded cylinder was then captured again with the light field camera. While the measurement with the laser distance sensor took several hours to complete, the light field camera only needed several seconds to capture each plane. Fig. 4 shows the surface profile as measured with the light field camera. The image was composed of nine individual images, which resulted in slight discontinuities at the interfaces. These can be explained by small distortions at the edges of the images as well as by slight inaccuracies which occurred during the realignment of the camera. Compared to the measurements from the laser distance sensor, a much higher spatial resolution was obtained, so that even very small structures can be identified. In order to directly compare both measurement techniques, three different slices at the positions y/L = 0, y/L = -0.5 and y/L = -0.8 were analyzed respectively. The positions are marked with white lines on the measured surface profile in Fig. 4.

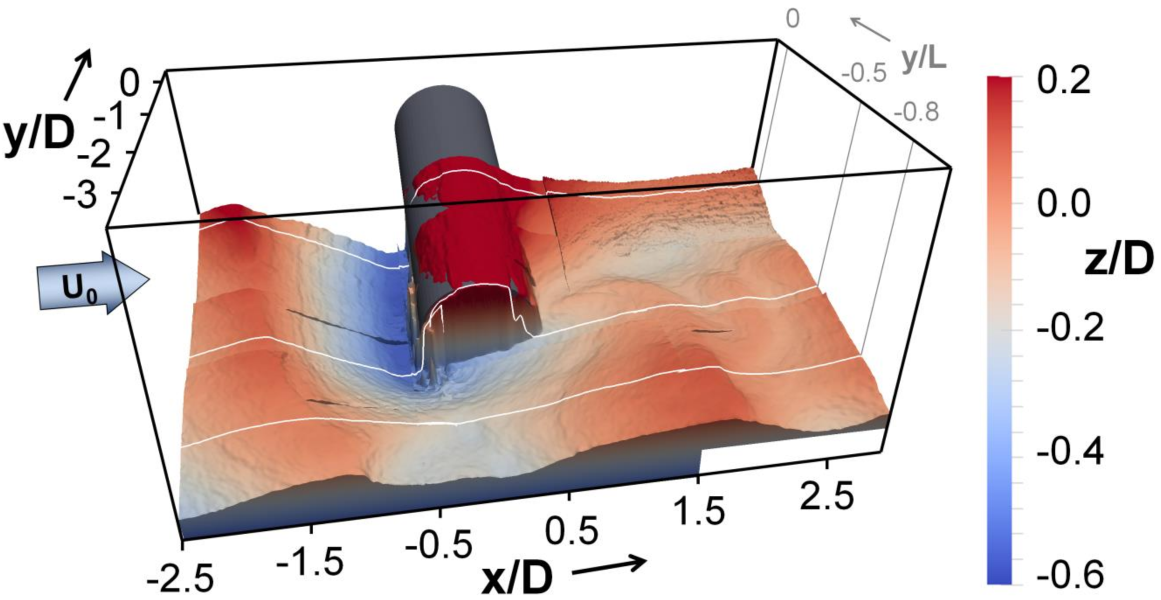


Fig. 4: Surface profile as measured with the light field camera.

Fig. 5 – 7 show the comparison of the measured profiles at the various positions. The profiles captured by the light field camera are shown in red and the measurements from the laser distance profile are represented by the black curves. The cylinder appears to be elliptical due to the overstretched vertical axis.

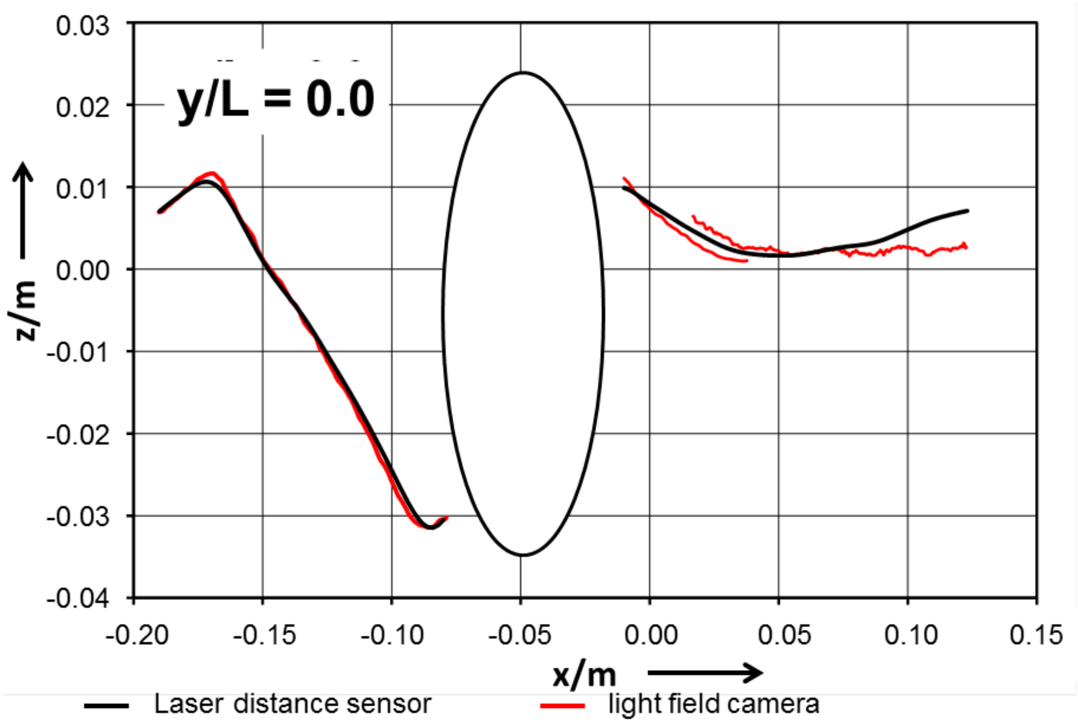


Fig. 5: Surface profile in the centre of the cylinder.

It is apparent that both measurements agree well and differ by less than 2 mm. The only exception is the area around 0.01 m < x < 0.12 m in Fig. 5, where insufficient lighting lead to higher uncertainties in the measurement data of the light field camera. Thus, the accuracy of the measurements are evidently strongly dependent on the quality of the lighting.

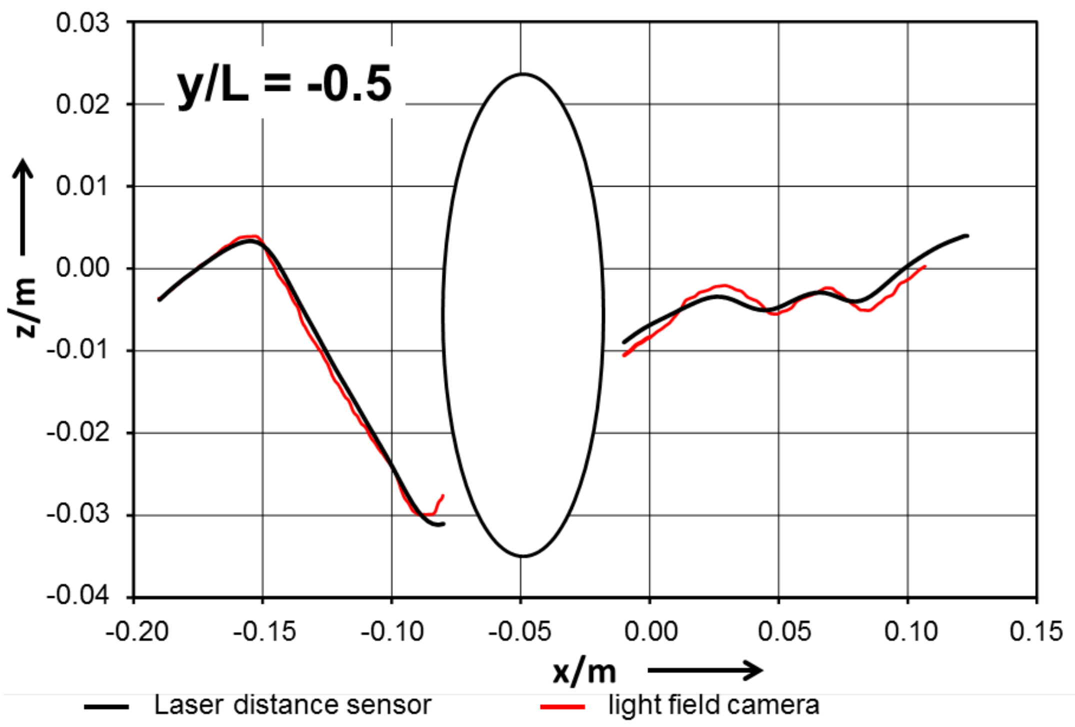


Fig. 6: Surface profile at the end of the cylinder.

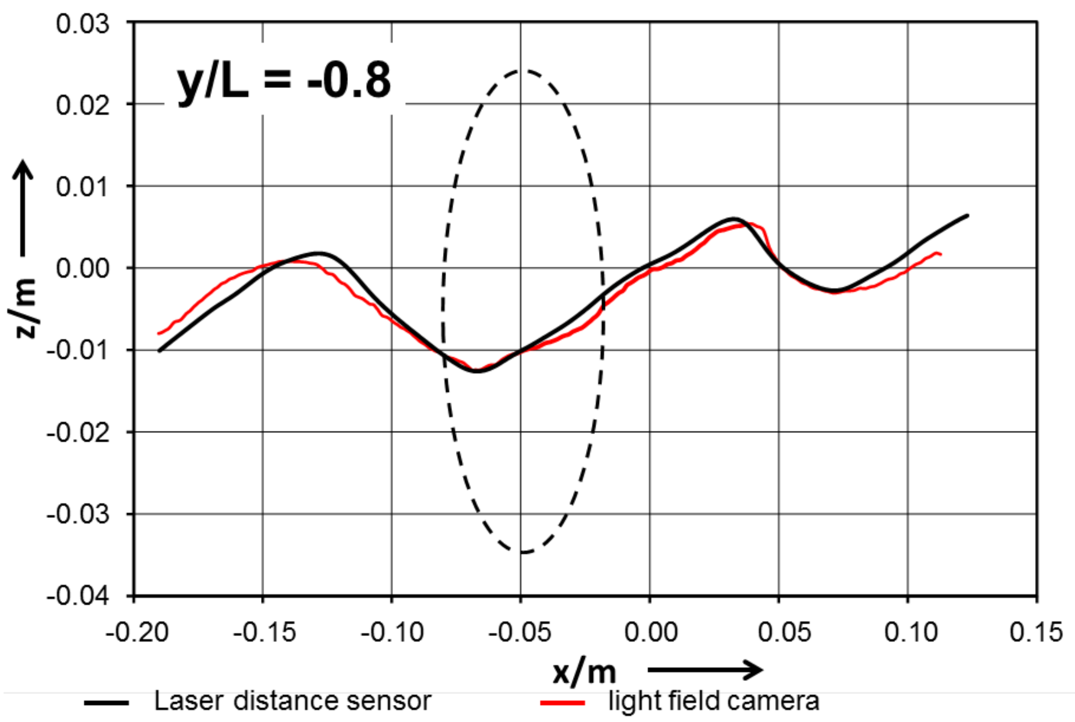


Fig. 7: Surface profile sideways of the cylinder.

When comparing the measured profiles, one must keep in mind though, that the data from the laser distance sensor had to be strongly smoothed to minimize the difference between the measurement points due to the inaccuracy of the technique. This then also resulted in a reduction of the statistical error.

Fig. 6 shows the surface profiles at the end of the cylinder, which are also in excellent agreement with each other. There are slight differences downstream of the cylinder, though these can partially be explained by the errors made during the positioning of the camera in the y-direction. Due to the large gradients in the surface height in both directions, the vertical error scales with the horizontal one. This effect also occurs sideways of the cylinder (Fig. 7).

In conclusion, the accuracy of the surface profile measurements with both the light field camera and the laser distance sensor are generally comparable, though the measurement time is significantly smaller with the light field camera. In addition, the surface data from the light field camera is easily accessible for post-processing. It was also shown though, that the accuracy of the measurements strongly depends on the quality of the lighting, the exact alignment and the accurate positioning of the camera.

Velocity measurements

In order to obtain an averaged velocity field, 752 double images (1504 individual images) were captured with the supplied software at a rate of 4 Hz.

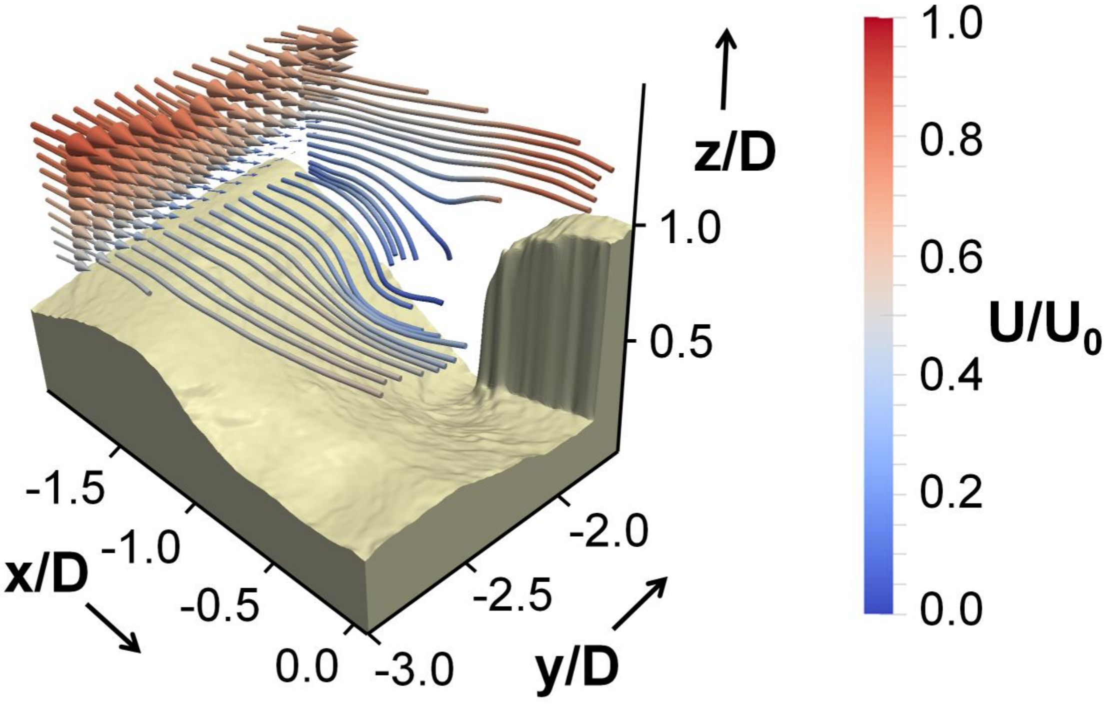


Fig. 8: Velocity field in front of the cylinder, visualized with velocity vectors and streamlines.

The velocity fields that were obtained via particle tracking were subsequently averaged on a uniform grid. Fig. 8 shows an excerpt of the resulting velocity fields, with the flow direction depicted by vectors. The streamlines indicate the structures which occur on average. Thus, the acceleration of the flow beside the cylinder and the flow around the edge of the cylinder becomes apparent. A validation of the measured velocities is to be carried out in future work.

The resulting three-dimensional three-component velocity field, together with the three-dimensional surface profile, allows for a detailed analysis and description of the flow structures in combination with the surface structure. Fig. 9, for example, shows the velocity field at around y/L = 0.45. A decreasing surface profile is visible in the area upstream of the cylinder, as well as a stagnation point right in front of the cylinder. The top of the resulting and rather extended horse shoe vortex in the ditch in front of the cylinder is visible in the measured velocity field. This horse shoe vortex was previously also found in wind channel measurements, though there it was not quite as extended. The horse shoe vortex which is to be expected here is marked schematically in Fig. 8 by the red arrow.

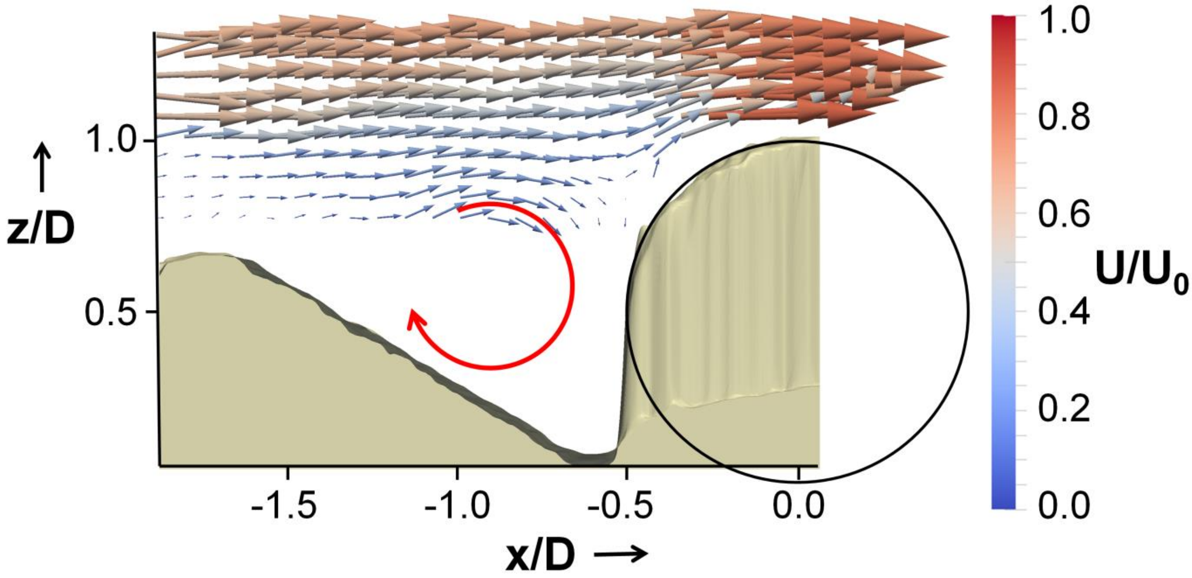


Fig. 9: Velocity field in front of the cylinder.

Summary and Outlook

The light field camera system was successfully tested and found to be suitable for the desired task. It was able to simultaneously measure 3D surface profiles as well as 3D-3C velocity fields and is therefore ideal for analyzing the interaction between the flow and a solid structure. Future work will address the temporal development of the measured structures. In addition, the measurement system is to be extended by additional possibilities for the calibration of the camera as well as for the analysis and export of the data. A system which allows for the control of the laser is also in development.

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